



AGH UNIVERSITY OF SCIENCE AND TECHNOLOGY
IM. STANISŁAWA STASZICA W KRAKOWIE
Faculty of Computer Science, Electronics and Telecommunications
DEPARTMENT OF ELECTRONICS

KE

ELECTRONIC DEVICES

Piotr Dziurdzia, Ph.D.

C-3, room 413; tel. 617-27-02, Piotr.Dziurdzia@agh.edu.pl



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PHYSICS OF SEMICONDUCTORS

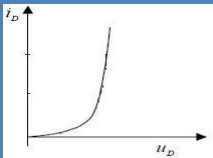
..... a journey in searching „carriers-conductors” of electrical current



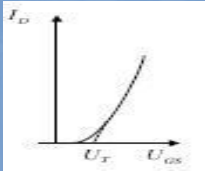
AFTER THIS CHAPTER WE SHOULD EASIER UNDERSTAND WHY:

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- in the junction semiconductor devices, the current-voltage characteristics are expressed by exponential relations ?



- in field effect semiconductor devices, the current-voltage characteristics are expressed by square relations ?



- in semiconductors, almost everything depends on temperature **T**, which sometimes is called as super parameter ?

$$R=f(U, I, T)$$



WHAT ARE WE GOING TO TALK ABOUT ?

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SEMICONDUCTOR MATERIALS

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THE CLASSIFICATION OF MATERIALS IN TERMS OF ELECTRICAL CONDUCTIVITY



resistivity

$$R = \rho \frac{l}{S} = \frac{1}{\sigma} \frac{l}{S}$$

conductivity

$$\rho [\Omega m]$$

$$\sigma \left[\frac{1}{\Omega m} \right]$$

$$\rho \sim \exp(-T)$$

$$\rho \sim T$$

INSULATORS

SEMICONDUCTORS

METALS

$$10E-6(\Omega m)E-1$$

$$10E+6(\Omega m)E-1$$

12 orders of magnitude!
(in room temperature)

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SEMICONDUCTOR MATERIALS

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Semiconductors - their important feature is that the conductivity may vary over a wide range due to changes in temperature, light or introduced dopants

III	IV	V
B	C	P
Al	Si	As
Ga	Ge	Sb
In	Sn	
III-V		
AlP		
AlAs		
GaP		
GaAs		
GaSb		

Elementary semiconductors

Complex semiconductors

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SILICON ATOM

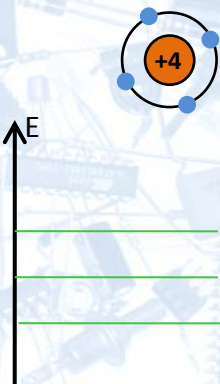
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According to Niels Bohr's theory of an isolated atom, the electrons can have strictly defined energy levels expressed in a quantum way:

atomic number of the element ($Z=14$) elementary charge of the electron ($1.6E-19C$) mass of the electron ($1.78E-31kg$)

$$E = \frac{Ze^4 m_e}{8n^2 h^2 \epsilon_0^2}$$

the number of electron shell Planck constant ($6.625E-34Js$) permittivity of the vacuum ($8.854E-12F/m$)



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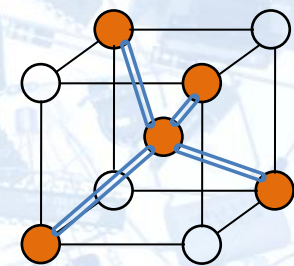
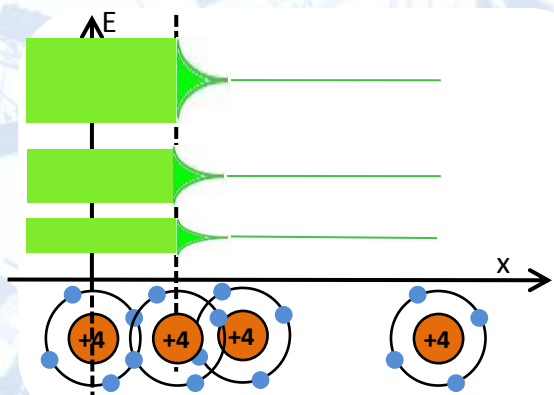
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SILICON ATOMS

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$1cm^3 Si \Rightarrow 10^{23} atoms$

If the band gap is: $1eV \Rightarrow 10^{-23} eV$ the distance between energy levels

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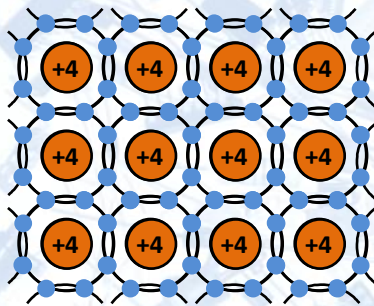


SEMICONDUCTOR MODELS INTRINSIC SEMICONDUCTORS

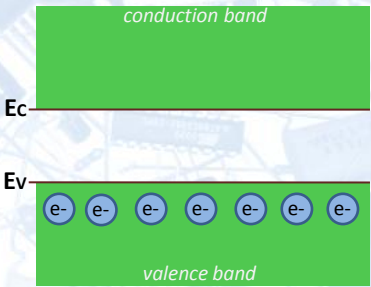
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in temperature $T=0K$

Two-dimensional model of a semiconductor from IV group



Energy band model



np. for Si $\rightarrow E_g=1.1eV$
for Ge $\rightarrow E_g=0.67eV$

$$E_c - E_v = E_g [eV]$$

energy gap

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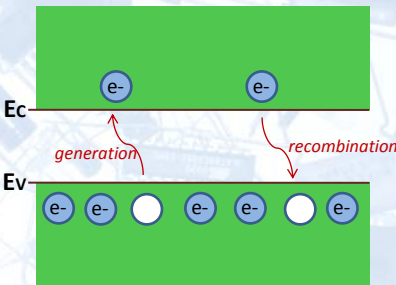
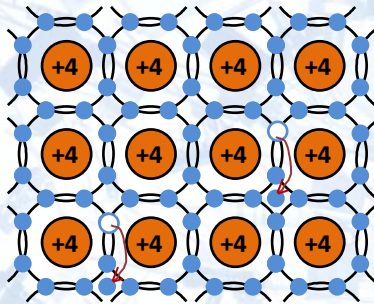
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SEMICONDUCTOR MODELS INTRINSIC SEMICONDUCTORS

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in temperature $T>0K$



Generation of electron-hole pairs can occur, for example under the influence of heat, light, radiation, ionization collision

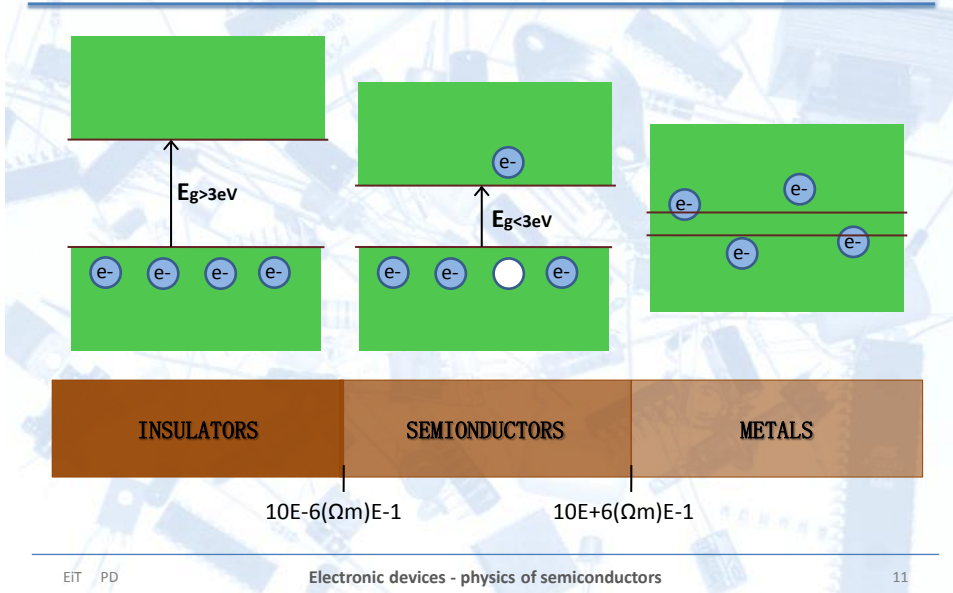
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BAND MODEL OF ELECTRIC CONDUCTION ^{KE}



STATISTICS IN SEMICONDUCTORS ^{KE}

What is the probability that an electron takes any energy state (level) E in absolute temperature T ?

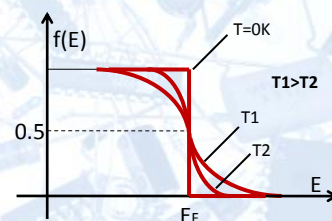
$$f(E) = \frac{1}{1 + e^{\frac{E - E_F}{kT}}}$$

Fermie-Dirac's function

$k = 8.62\text{E-}5\text{eV/K} = 1.38\text{E+}23\text{J/K}$ - Boltzmann's constant

What is E_F ?

$$f(E_F) = \frac{1}{1 + e^{\frac{E_F - E_F}{kT}}} = \frac{1}{2}$$



Energy state located at the Fermie's level can be taken by an electron with the probability of 0.5



STATISTICS IN SEMICONDUCTORS

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How can we determine the concentrations of electrons and holes in a semiconductor unit volume?

electrons

$$f_n(E) = \frac{1}{1 + e^{\frac{E-E_F}{kT}}}$$

$$\text{dla } |E-E_F| > 3kT \quad f_n(E) = e^{-\frac{E-E_F}{kT}}$$

The energy density of available energy states for electrons in the conduction band:

$$N_C(E) = \frac{4\pi(2m_e^*)^3}{h^3} \sqrt{E - E_C}$$

m_e^* — effective mass of the electron

$$n = \int_{E_C}^{\infty} N_C(E) f_n(E) dE = N_C e^{-\frac{E_C-E_F}{kT}}$$

$$N_C = 2 \left(\frac{2\pi m_e^* kT}{h^2} \right)^{\frac{3}{2}} \text{ — effective density of energy states in the conduction band}$$

holes

$$f_p(E) = \frac{1}{1 + e^{\frac{E_F-E}{kT}}}$$

$$\text{dla } |E_F-E| > 3kT \quad f_p(E) = e^{-\frac{E_F-E}{kT}}$$

The energy density of available energy states for electrons in the valence band:

$$N_V(E) = \frac{4\pi(2m_h^*)^3}{h^3} \sqrt{E_V - E}$$

m_h^* — effective mass of the hole

$$p = \int_{-\infty}^{E_V} N_V(E) f_p(E) dE = N_V e^{-\frac{E_F-E_V}{kT}}$$

$$N_V = 2 \left(\frac{2\pi m_h^* kT}{h^2} \right)^{\frac{3}{2}} \text{ — effective density of energy states in the valence band}$$

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STATISTICS IN SEMICONDUCTORS

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In intrinsic semiconductor:

$$n = p \equiv n_i$$

$$n_i(T) = \sqrt{np} = \sqrt{N_C N_V} e^{-\frac{E_C-E_V}{2kT}} = AT^{\frac{3}{2}} e^{-\frac{E_g}{2kT}}$$

$$A = \frac{4\pi}{h^2} (m_e^* m_h^*)^{\frac{3}{4}}$$

The law of mass:

$$n_i^2 = np$$

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CURIOSITY I

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In $1\text{cm}^3\text{Si}$ there is
 10^{23} atoms



In silicon the energy
gap is: $E_g = 1.1\text{eV}$





$$n_i(T) = AT^{\frac{3}{2}} e^{-\frac{E_g}{2kT}}$$

$$n_i(300\text{K}) = 1.5 \cdot 10^{10} \text{cm}^{-3}$$

so, in 1mm^3 we can find
15 millions of free electrons !!!
and the same number of holes ;))

If the average thermal energy $E_t = kT$ of an electron at room
temperature $T = 300\text{K}$, is
 $E_t = 0.025\text{eV}$, then how they can
overcome the energy gap?



Sufficient energy to overcome the energy gap in
silicon at room temperature has one electron to
 1.5×10^{13} atoms!!!



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CURIOSITY II

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What is the sensitivity of free electrons and holes
concentration changes in the intrinsic silicon at
the ambient temperature $T = 300\text{K}$?



we should calculate:

$$\gamma_i = \frac{dn_i}{n_i} = \frac{3}{2T} + \frac{E_g}{2kT^2}$$



after substituting the data
we obtain $\gamma_i(300\text{K}) = 8.3\%$



W. Janke, „Zjawiska termiczne w elementach i układach półprzewodnikowych”, WNT1992

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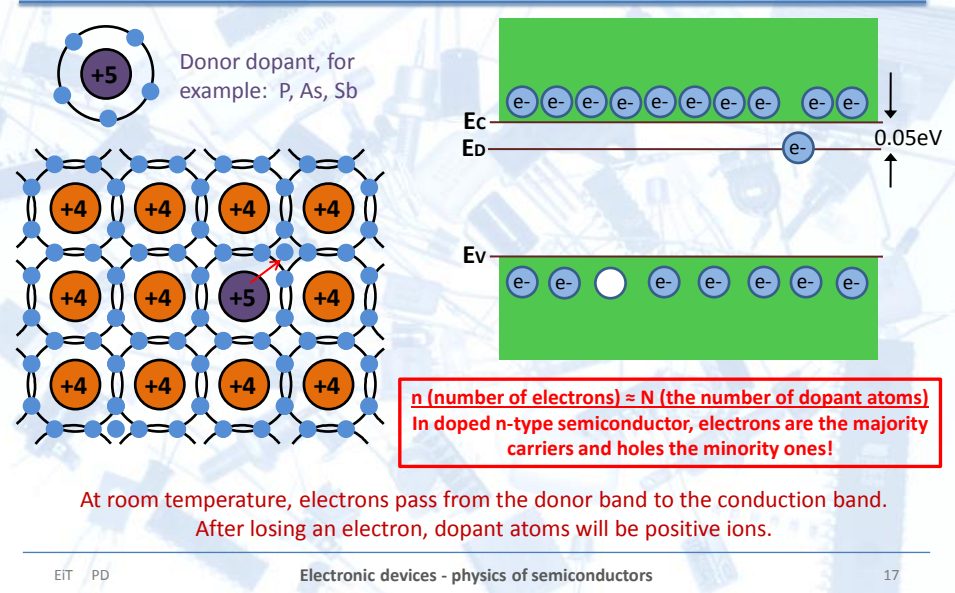
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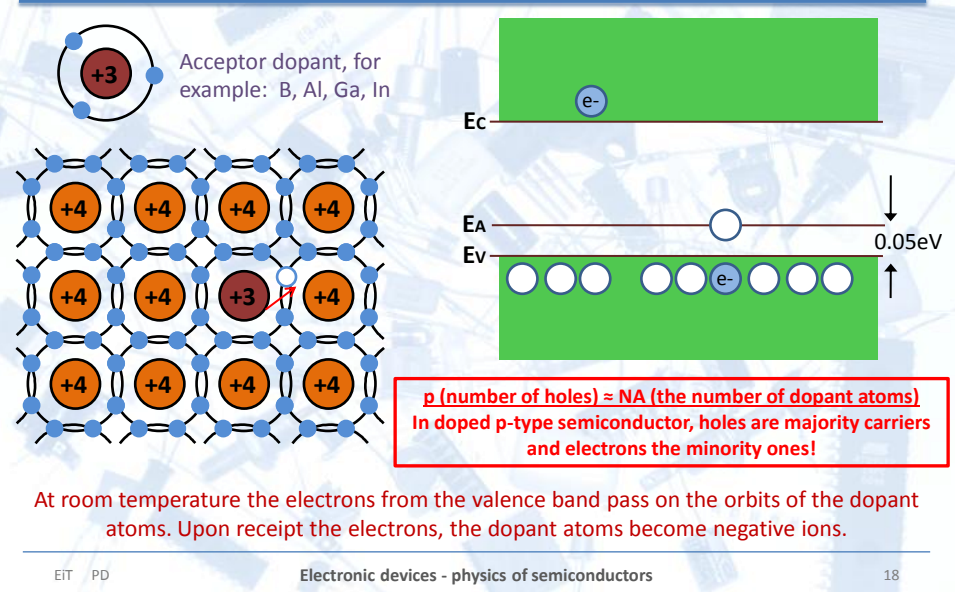
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DOPED SEMICONDUCTORS

KE





DOPED SEMICONDUCTORS

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Condition of electrical neutrality: the introduction of dopants into the semiconductor can not change the total charge load, which is in equilibrium and must be equal to zero.

$$N_D^+ - n + p - N_A^- = 0$$

From the law of mass we can determine the concentration of free carriers for a known concentration of dopants:

for donor semiconductors: $n_n \approx N_D$ $p_n \approx \frac{n_i^2}{N_D}$

for acceptor semiconductors: $p_p \approx N_A$ $n_p \approx \frac{n_i^2}{N_A}$

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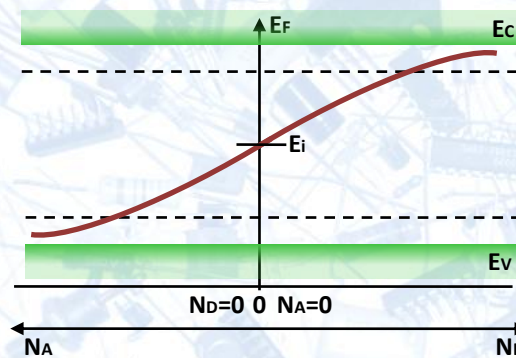
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DOPED SEMICONDUCTORS

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The introduction of dopants causes changes in the position of the Fermi level



The position of the Fermi level is also a function of temperature

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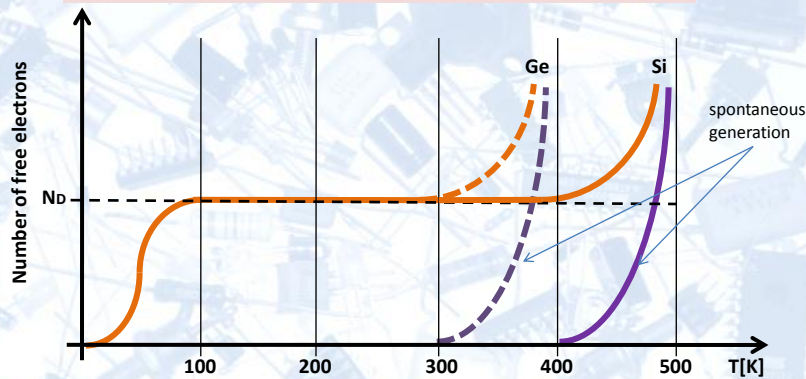
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DOPED SEMICONDUCTORS

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Temperature dependence of carrier concentration in the donor semiconductor



Doping causes a stabilization of a number of carriers at a relatively high temperature range!

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CURIOSITY III

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What will be the effect of doping: 1 arsenic atom (As +5) to 1 million silicon atoms (Si +4) ???



In $1\text{cm}^3\text{Si}$ there is 10^{23} atoms



After doping we obtain $\frac{10^{23}}{10^6} = 10^{17}$ arsenic atoms and the same number of free electrons in room temperature!



The introduction of impurities (dopants) with a density of one particle per million (ppm) may change intrinsic silicon in a sample of excellent conducting material !!!

$$\begin{aligned}\rho_{\text{Si}} &= 2 \cdot 10^3 [\Omega\text{m}] \\ \rho_{\text{Si-As}} &= 2 \cdot 10^{-3} [\Omega\text{m}] \\ \frac{\rho_{\text{Si}}}{\rho_{\text{Si-As}}} &= \frac{2 \cdot 10^3}{2 \cdot 10^{-3}} = 10^6\end{aligned}$$



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ELECTRICAL CONDUCTIVITY IN SEMICONDUCTORS

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In the absence of an electric field the electrons follow a chaotic motion. At room temperature, the average thermal velocity is about

$$2 \cdot 10^5 \left[\frac{m}{s} \right]$$

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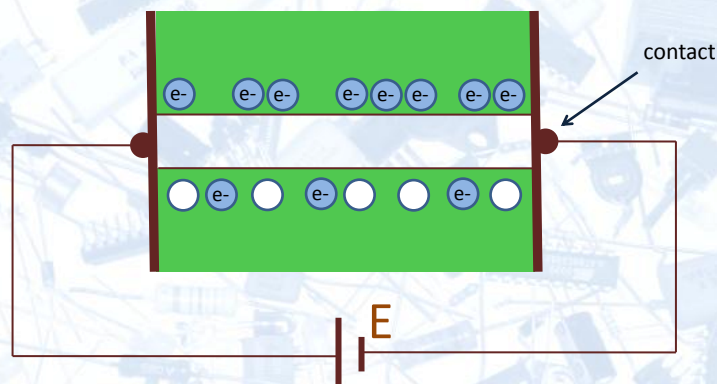
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ELECTRICAL CONDUCTIVITY IN SEMICONDUCTORS

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Upon application of an external electric field, there appears a uniform movement of electrons – drifting the free carriers in an electric field. At room temperature the velocity of transport is about ...

$$10^{-4} - 10^{-3} \left[\frac{m}{s} \right]$$

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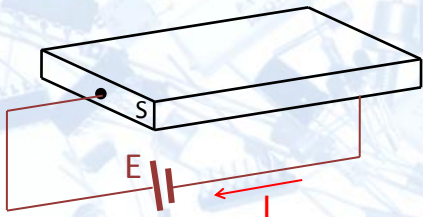
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ELECTRICAL CONDUCTIVITY IN SEMICONDUCTORS

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Drift current density – 1-D considerations



electric charge flowing through a surface S during time dt:

$$dQ = n \cdot e \cdot v_n \cdot dt \cdot S + p \cdot e \cdot v_p \cdot dt \cdot S$$

current density:

$$J = \frac{1}{S} \frac{dQ}{dt}$$

$$J = n \cdot e \cdot v_n + p \cdot e \cdot v_p$$

according to Ohm's Law:

$$J = \sigma E$$

so, the conductivity:

$$\sigma = e(n\mu_n + p\mu_p)$$

$$v_n = \mu_n E \text{ mobility of electrons}$$

$$v_p = \mu_p E \text{ mobility of holes}$$

for silicon $\mu_n \approx 3\mu_p$



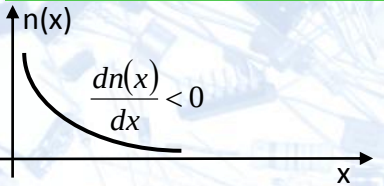
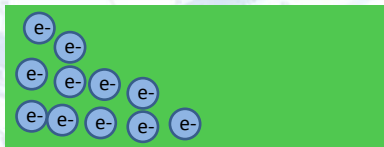
ELECTRICAL CONDUCTIVITY IN SEMICONDUCTORS

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Diffusion of electric carriers

Diffusion currents occur in the states of imbalance in a semiconductor material where the carrier concentration becomes heterogeneous.

The electric carriers move from an area of high concentration to the area of low concentration, until the distribution of free carriers is homogenous.



$$J = -qD \frac{dn(x)}{dx}$$

diffusion coefficient



ELECTRICAL CONDUCTIVITY IN SEMICONDUCTORS

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$$\frac{D_n}{\mu_n} = \frac{D_p}{\mu_p} = \frac{kT}{e} = U_T = 26mV (T = 300K)$$

$$J_n = n \cdot e \cdot \mu_n \cdot E + e \cdot D_n \frac{dn(x)}{dx}$$
$$J_p = p \cdot e \cdot \mu_p \cdot E + e \cdot D_p \frac{dp(x)}{dx}$$

drift

diffusion